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Successful Appraisal Of a Very Low Permeability Fractured Carbonate Reservoir, Bab Member, Offshore Abu Dhabi

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Abstract

The Bab member, the topmost unit of the Thamama reservoir, has historically shown very poor permeability and due to its tightness and surface constraints, it has never been considered as a primary objective. This paper describes how this reservoir was reassessed and tested, unlocking significant additional reserves in line with ADNOC's aspiration to increase hydrocarbon production.

The primary objective was to select a well with comprehensive core data and high-quality logs for this operation. Following that, a detailed re-evaluation of the sedimentological description and reservoir properties of the Bab member was conducted using previous learnings to pinpoint potential contributing intervals. Subsequently, in collaboration with the operations and reservoir management teams, a perforation sequence was meticulously designed to identify these contributing intervals and assess vertical permeabilities. Finally, PBU and PLT were conducted to enhance our understanding of the reservoir's permeabilities and contributing reservoir thickness.

The sedimentological reassessment and reservoir characterization identified three distinct units within the Bab member. The upper unit comprises laminated, organic-rich wackestone, the middle unit consists of foraminiferal, skeletal wackestone, and the lower unit like the upper with laminated, organic-rich wackestone. Notably, the middle unit emerged as the sole reservoir candidate.

Following these findings, a two-stage perforation strategy was devised. The initial stage targeted the lower unit and the base of the middle unit, resulting in an acceptable production rate. The subsequent stage focused solely on the middle unit, yielding a modest production increment, indicative of high vertical permeabilities.

Production Logging Tool (PLT) data confirmed that production was solely from the middle unit, while the lower unit functioned as a seal, corroborated by core data. Well test analysis revealed permeabilities thickness (KH) five times greater than suggested by core data, hinting at the existence of diffused fractures validated through core descriptions.

This operation has revealed and tapped into the true potential of the Bab member within our field. Moreover, based on paleogeographic mappings, it is likely that the Bab member, exhibiting similar characteristics, is present in other offshore fields. Consequently, analogous operations can be strategized to unlock further potential reservoirs. This discovery significantly contributes to ADNOC's goal of accelerating production.

Introduction

The Lower Cretaceous in the studied field consists of a fractured carbonate reservoir situated at the crest of the field, exhibiting low-quality matrix behaviour on the flanks. Oil production is derived from two intervals, R1/R2 and R3/R4 (5). These intervals comprise limestone with relatively high porosity (greater than 20%) and poor matrix permeability (ranging between 2 and 10 millidarcies). The reservoir units are interspersed with denser, more fractured layers. The average thickness of each main sub-unit ranges from 40 to 100 feet (1).

Interpretation of 3D seismic data indicates that the overall structure of the field is a broad, slightly elongated dome-oriented NE-SW, with gently dipping flanks and numerous NW-striking normal faults exhibiting small throws cutting through the reservoir. These faults are oriented perpendicular to the main current horizontal stress, suggesting they are closed and do not significantly contribute to fluid flow (1). Analysis of core data, image logs and seismic attributes reveals the existence of a major set of open fractures oriented N10 to 30E across the crest of the structure. The contribution of these fractures to fluid flow is well documented through production data (8).

The Bab member, the uppermost unit of the Thamama reservoir, has historically demonstrated very poor permeability and due to its intrinsic tightness, it has not been considered as a primary objective. Sedimentological reassessment and reservoir characterization have identified three distinct units within the Bab member: the upper unit comprising laminated, organic-rich wackestone; the middle unit consisting of foraminiferal, skeletal wackestone; and the lower unit, similar to the upper, with laminated, organic-rich wackestone. Notably, the middle unit emerged as the sole viable reservoir candidate due to the presence of open fractures.

This paper outlines the reassessment and testing of this reservoir, leading to the unlocking of significant additional reserves, aligning with ADNOC's goal of increasing hydrocarbon production.

Field Structure

The studied offshore field structure is circular structure located on a basement high that strikes at about NNE-SSW. This later started to develop during the Permian NE-SW extension that continued throughout much of the Mesozoic (1), (2), (3), (9). The NW-SE normal faults in the Lower Cretaceous were caused by the mesozoique NE-SW extension with the stratigraphic position of these faults indicating they developed in the Early to Middle Cretaceous (1), (2), (3), (9) Figure 1. These faults are approximately parallel to each other and extend out of the studied offshore field fold suggesting they are related to regional extension rather than to doming. These normal faults are associated with dominantly closed and commonly mineralized fractures that strike NW-SE Figure 2. The current regional stress system is related to the development of the Zagros Mountains with the maximum horizontal compressive stress oriented around 30° (NW-SE). Such a stress regime has created a dominant set of fractures which are open and strike around 30° (5). The studied offshore field fold continues to develop in this stress system Figure 2.

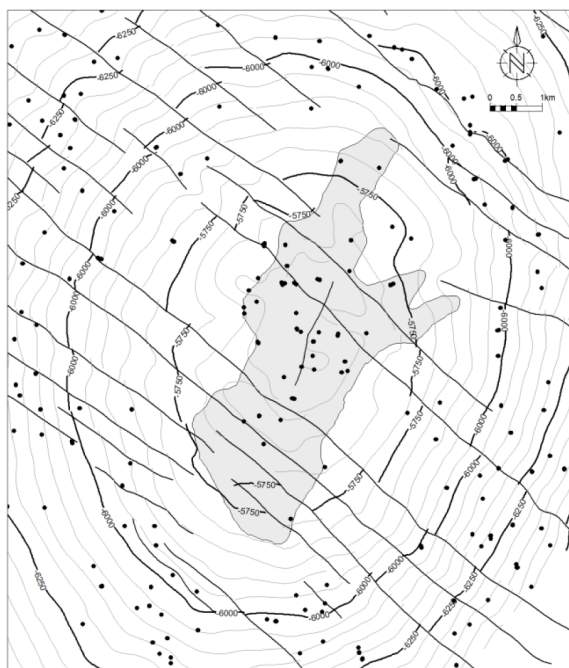


Figure 1—Structural map of the lower cretaceous in our field. The faults are approximately parallel to each other and extend out of the fold suggesting they are related to regional extension rather than to doming.

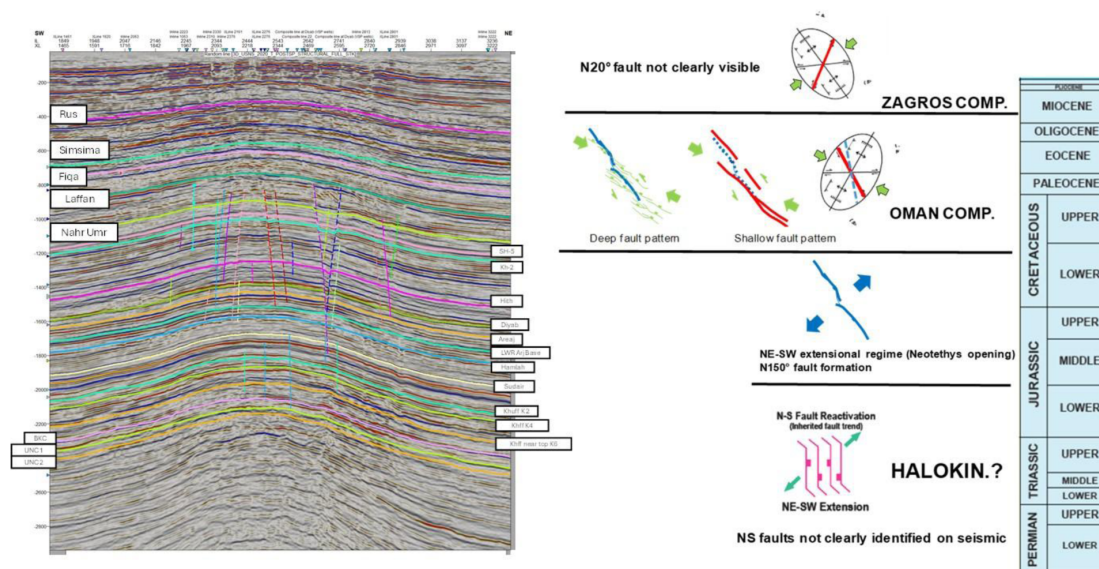


Figure 2—Structural architecture of our field and main geological events which have controlled its geological evolution.

Stratigraphy of Lower Cretaceous Group

In the offshore areas of UAE, the Lower Cretaceous shows a well-developed alternation of porous, cleaner zones with less porous, dense ones. The thickness of the porous and dense zones varies little over tens of kilometres, therefore a 'layer cake' or Cyclo-stratigraphic approach has traditionally been used to zone the reservoir. Several sequence stratigraphic models were applied by different authors (1), (2), (3), (4), (9).

In the Offshore Field, the focus is on the upper part of the Lower Cretaceous reservoir, and it involves R1/R2, D1 (Aptian) and R3/R4 (Barremian) Formations Figure 3. In our paper we will focus on the top part of the Aptian R1.

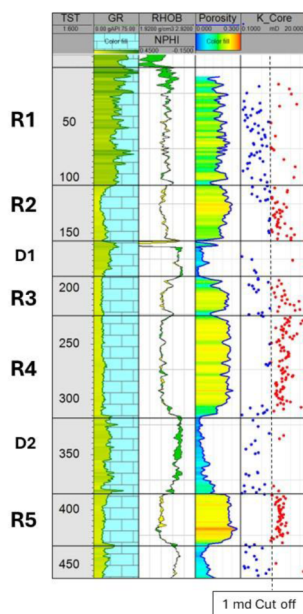


Figure 3—Stratigraphic column and main reservoir characteristics of the upper part of the Lower Cretaceous formation in our field. It shows a well-developed alternation of porous, cleaner zones with less porous, dense ones. Matrix permeabilities are generally low (typically ~0.2-4 mD).

Bab Member sedimentology

The Bab Member is characterized by deep-water facies deposited within a restricted range. It serves as the condensed, distal equivalent to the upper section of the underlying Shu'aiba Formation's platform margin shoal carbonates (7). This member encompasses one third-order sequence and three fourth-order sequences (Apt 2, 3, and 4). The sequence Apt 5 represents the LST of a subsequent super-sequence extending into the Nahr Umr Formation [Figure 4](#).

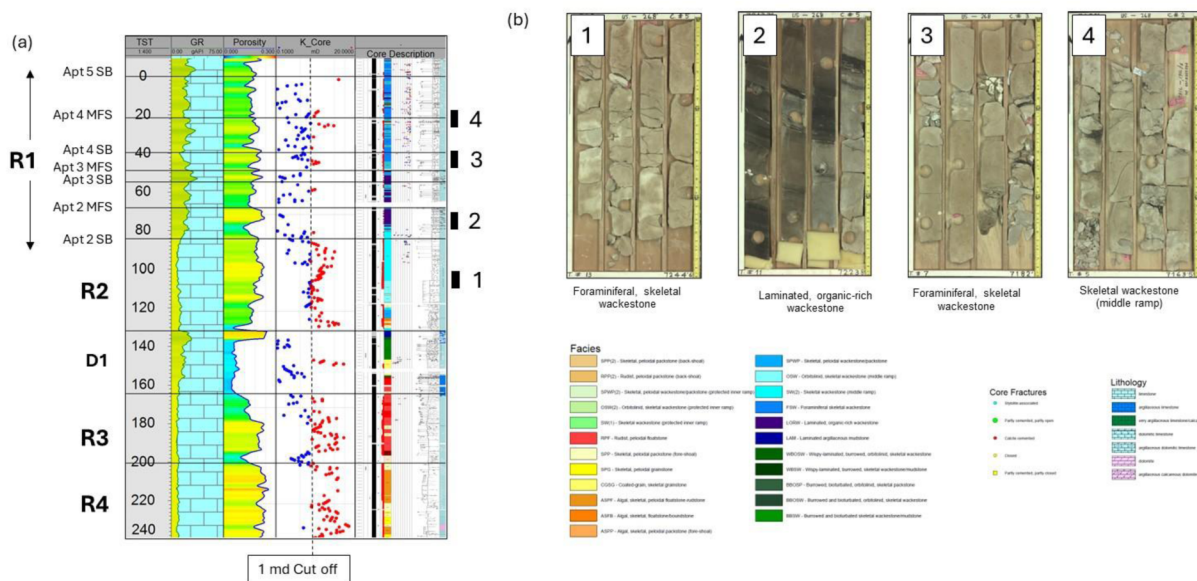


Figure 4—(a) Stratigraphic column, core description and main reservoir characteristics of the Bab Member which encompasses one third-order sequence and three fourth-order sequences (Apt 2, 3, and 4). The sequence Apt 5 represents the LST of a subsequent super sequence extending into the Nahr Umr Formation. (b) core photos of the different fourth-order sequences of the Bab member.

- The Apt 2 TST is distinguished by its high organic matter content, evident through distinctive wireline log signatures and typically low grain densities. Core samples reveal black, finely laminated, carbonaceous lime mudstones and wackestones, interspersed with burrowed skeletal lime wackestones/floatstones rich in planktonic foraminifera. This sequence was deposited during an oceanic anoxic event, which caused the oxygen minimum zone to impinge on the Bab Intra-shelf Basin (7).
- The Apt 3 and 4 consist of lime mudstone/wackestone, with grain types predominantly composed of whole thin-shelled bivalves and gastropods, and some visible planktonic foraminifera in thin section. These sequences are characterized by wispy-laminated, bioturbated, and burrowed structures throughout.
- The Apt 5 LST is notably more argillaceous than the underlying sequences, as indicated by the NPHI curve plotting to the left of the RHOB curve and a relatively high GR response on wireline logs. The facies are similar to those in the Apt TST, though less carbonaceous and more argillaceous. Regional data suggest this sequence was deposited on prograding ramps during continued sea-level fall; thus, it is interpreted as an LST (7). The clay abundance in this sequence is seen as a distal manifestation of the Burgan Delta's advancement in Kuwait and Iraq, which eventually led to the deposition of the Nahr Umr Formation in this area.

In the Bab Member, vertical variations in NPHI and RHOB are generally subtle, except in the lower part of the Bab member, where high kerogen content causes a characteristic bow-shaped increase in NPHI, resistivity, and DT, alongside a decrease in RHOB and the grain densities measured in cores [Figure 3](#). The top of the Bab Member is challenging to pick consistently due to the rarity of useful wireline log data across the interval. [Figure 3](#).

Bab Member reservoir quality

Diagenetic alterations have significantly impacted the Thamama sediments, with a more pronounced effect on the denser zones compared to the porous zones. Key diagenetic processes influencing reservoir quality include the dissolution of unstable grains, calcite cementation, and both mechanical and chemical compaction. The Bab Member's preserved porosity is primarily microporosity associated with micritic grains and matrix, resulting in generally low matrix permeabilities ranging from approximately 0.1 to 2 millidarcies [Figure 4](#). Reservoir quality exhibits strong stratigraphic control, primarily due to the original detrital clay content dictating the extent of diagenetic overprint. Despite having moderate porosities (around 20-24%), the matrix permeabilities remain low (~0.1-2 millidarcies). Production test interpretations underscore the critical role of open fractures in enhancing reservoir performance.

Thamama fractures characterization

Fracture identification in the Lower Cretaceous reservoirs was carried out utilizing image logs, leading to a structural interpretation of 18 wells equipped with these logs. The dataset identified four distinct fracture sets: three conductive and one resistive [Figure 5](#). Set-1, with a strike orientation of N20-40° ($\pm 10^\circ$), represents the dominant fracture set, while the secondary sets exhibit varied strike orientations (1). Vertically, Set-1 is prevalent across all reservoirs, with Set-2 gaining significance in the R2 and R3 reservoirs, and Set-3 becoming more apparent in R2, R3, and R4 [Figure 5](#).

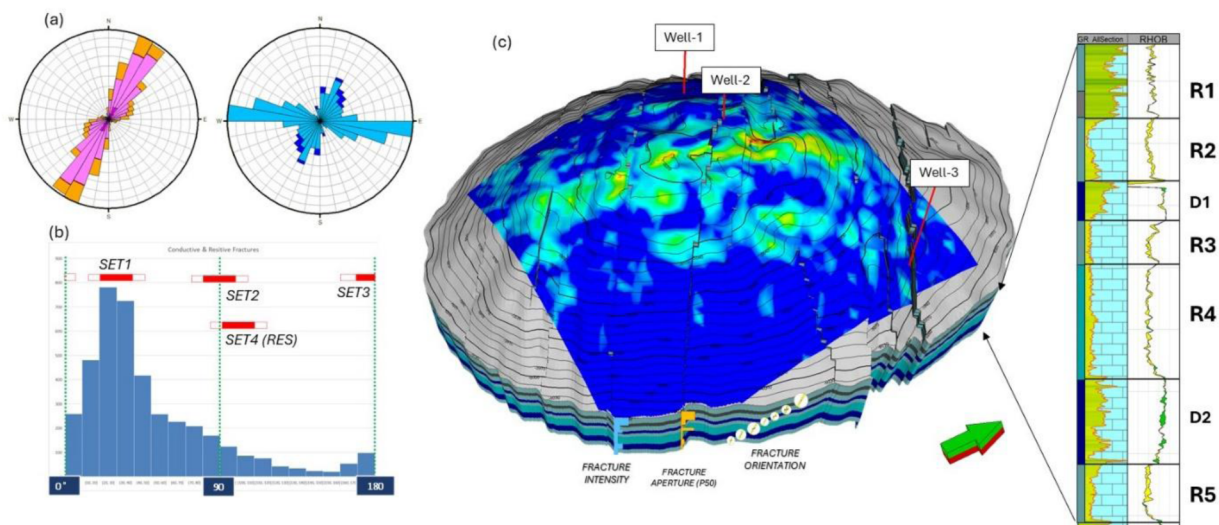


Figure 5—(a) Structural interpretation of 18 wells with image logs. The identified fracture dataset shows four distinct sets, three sets are conductive fractures, and one set is resistive fractures. (b) Set-1 includes fractures strike orientation N20-40 (+/- 10 degrees) represents the main and predominant fracture set. (c) 3D diagram showing the vertical distribution of fracture. Set-1 is predominant in all reservoirs, Set-2 is important in the R2 and R3 reservoirs, Set-3 is more evident in R2, R3 and R4.

Dynamic data provided further evidence of fractures. Pressure Transient Analysis (PTA) revealed significant discrepancies between the permeability height (KH) derived from logs and that obtained from well tests, indicating the presence of fractures Figure 6. PTA analysis also highlighted differences between wells situated on the flanks and those on the crest:

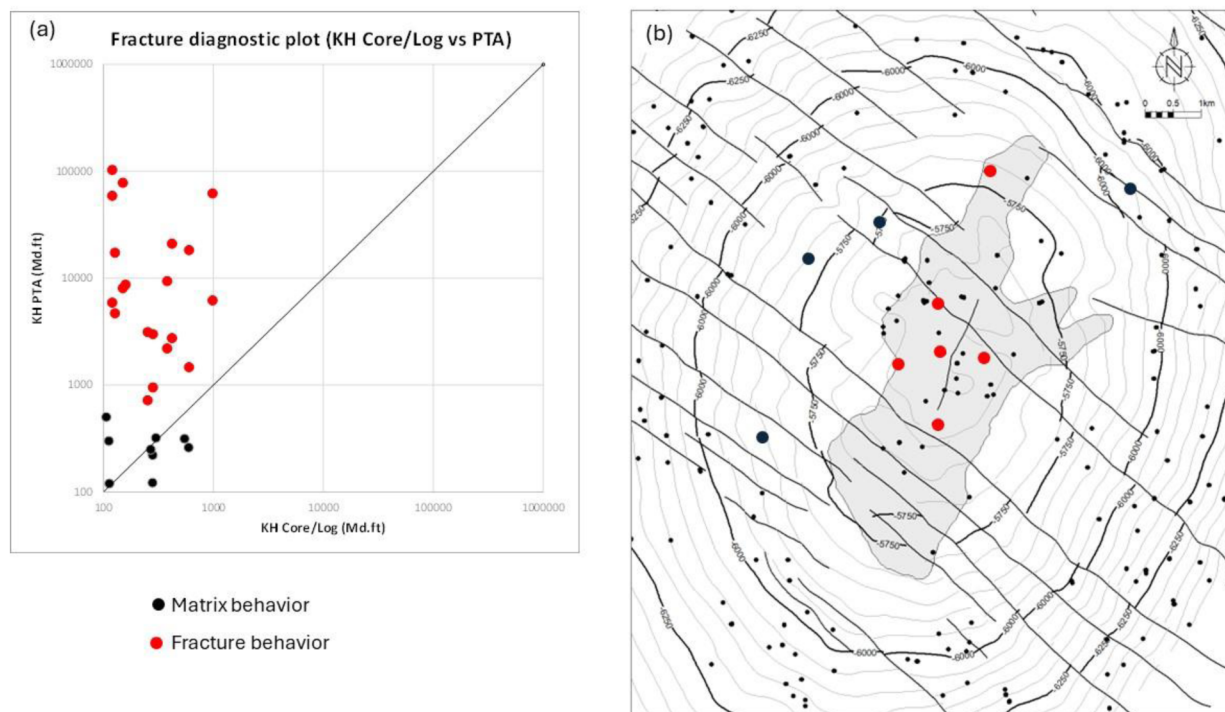


Figure 6—(a) The analysis of PTA shows significant differences between the KH from logs when compared to KH from well test indicating the presence of fractures. (b) The analysis of the same PTA shows significant differences between the wells on the flanks compared with those on the crest, indicating the presence of fractures in the crestal part of the field.

- Short-duration (approximately one day) tests on flank wells suggested hydrocarbon flow primarily through the matrix, with occasional short, isolated, open fractures.

- Most tests on crestal wells exhibited two stages: initial hydrocarbon flow dominated by a diffuse fracture network with conductivity in the range of thousands of millidarcy-feet (mD.ft), significantly higher than expected from the matrix; subsequently, conductivity increased to tens of thousands of millidarcy-feet (mD.ft), indicating an extension into a network of large open fractures over time.
- Some crestal wells displayed a single stabilization phase with conductivity in the tens of thousands of millidarcy-feet (mD.ft), suggesting penetration into a large open fracture network.

Bab member Fracture characterisation

To enhance fracture characterization within the Thamama reservoirs, two wells were strategically selected based on their structural positioning. The first well is situated within a fault block, distanced from the primary NW-SE fault lines, while the second well intersects these faults. Borehole imaging (BHI) interpretation of the two wells reveals that the majority of fractures are concentrated in the upper part of the Thamama reservoirs (Bab Member R1) [Figure 7](#).

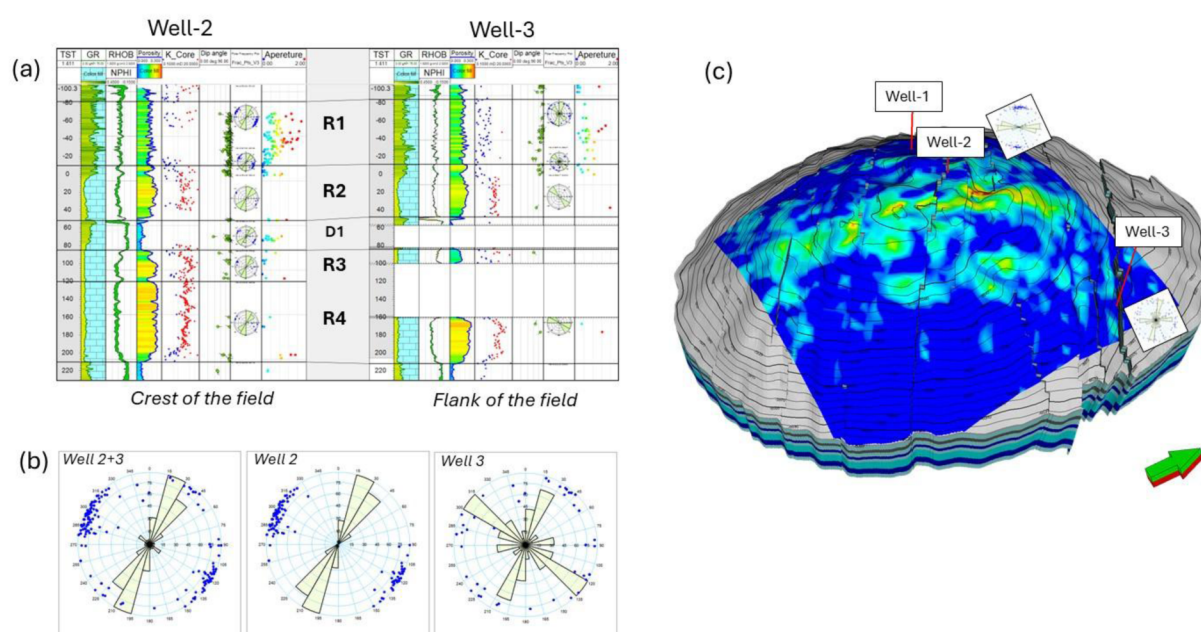


Figure 7—(a) Borehole imaging (BHI) interpretation of two wells reveals that the majority of fractures are concentrated in the upper part of the Thamama reservoirs (Bab Member R1) the analysis of fracture apertures reveals that the lower section of the Bab Member, exhibits minimal aperture values. (b) Rose diagram showing that both wells exhibit a consistent NE-SW fracture trend, Well-2 demonstrates a higher concentration of fractures, Well-3 displays a wider dispersion of fractures, reflecting local secondary fracture sets aligned with the NW-SE faults. (3) 3D diagram showing the decrease of fracture intensity from the field crest towards the flank.

Both wells exhibit a consistent NE-SW fracture trend, aligning with the prevailing regional stress regime and paralleling the main fracture corridors identified through seismic inversion in the crestal area of the field. Well-2 demonstrates a higher concentration of fractures, indicative of a more stable or less disturbed structural setting, as it is located centrally within a fault block, away from the major NW-SE faults. Conversely, Well-3 displays a wider dispersion of fractures, reflecting local secondary fracture sets aligned with the NW-SE faults [Figure 7](#).

Borehole imaging (BHI) interpretation indicates a decrease in fracture intensity and aperture from the field crest towards the flank, suggesting that the fractures are associated with geological folding. Additionally, analysis of fracture apertures reveals that the lower section of the Bab Member, corresponding to laminated, organic-rich wackestone, exhibits minimal aperture values, indicating nearly closed fractures.

This suggests that this interval will not contribute to flow, effectively isolating the middle part of the Bab Member from the R2 reservoir [Figure 7](#).

Bab member dynamic performance

Based on the sedimentological and fracture characterization of the Bab member, this later was perforated in two steps to validate the reservoir characteristics described previously. The first step targeted organic-rich wackestone with low matrix permeability and minimal fracture aperture, confirming its poor reservoir quality. The second step focused on skeletal wackestone (middle ramp) exhibiting open fractures. This sequential perforation aimed to assess the dynamic behaviour of each interval, ensuring the effective stimulation and optimization of hydrocarbon recovery from the fractured zones.

Well Intervention Summary

The objective of the rig-less intervention was originally to perform pre-rig securing given that both strings were unable to flow against the sea line pressure. However, following a multidisciplinary review of the well, the team had identified a unique opportunity to appraise Bab member through an innovative approach. Historically, it was challenging to appraise Bab member individually as it was often perforated in the short string and commingled with other layers, with no opportunity to run a PLT to assess the production profile due to inaccessibility [Figure 8](#).

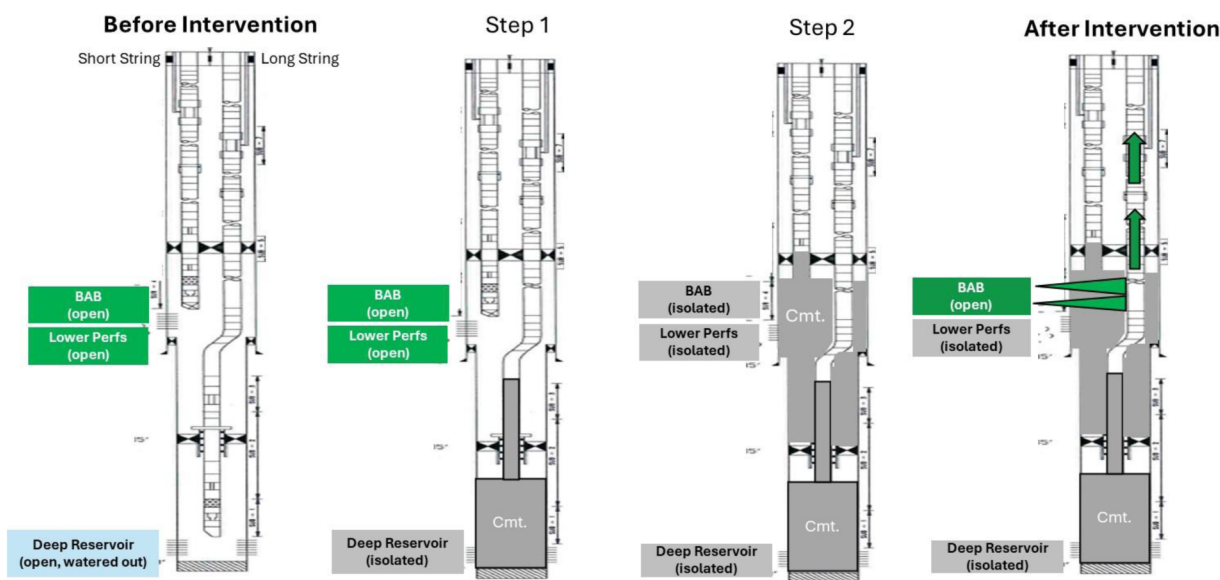


Figure 8—Well Schematic before and after intervention including the different steps of isolating existing perfs, followed by selective perforation of the upper body only

The challenging well configuration meant that the intervention plan had to include isolating the existing perforations in long and short string using coiled tubing cement plugs, followed by selective through-tubing perforation and stimulation of Bab member only. The success of the operation was contingent on performing a successful cement job to ensure contribution only from the top bodies planned to be reperforated. [Figure 8](#).

The cement job was successful; the well was displaced to diesel and Bab member was perforated in two steps. Initially, the well was unable to flow without stimulation, but following the stimulation job using HCl acid, productivity had improved allowing the well to start flowing. The well was tested and confirmed to be flowing with adequate oil rate and Well Head Flowing Pressure. The change in well performance (Well Head Flowing Pressure, Oil Rate) compared with pre-isolation flow tests re-confirmed successful isolation of the lower zones.

Production Logging Tool interpretation

Following the perforation of the well and the initiation of production, a Production Logging Tool (PLT) was deployed to evaluate the vertical dynamic behaviour of the Bab member. The PLT results indicated that the lower section of the Bab member, characterized by organic-rich wackestone with low matrix permeability and minimal fracture aperture, did not contribute to the flow, as anticipated. Conversely, all production originated from the skeletal wackestone (middle ramp) zone Figure 9. The PLT also demonstrated uniform contribution across this zone, without the presence of high-permeability streaks. Despite the low permeability observed in core samples, this interval contributed to production due to the presence of fractures, which were well-identified and described in image logs and core Figure 10 and Figure 11.

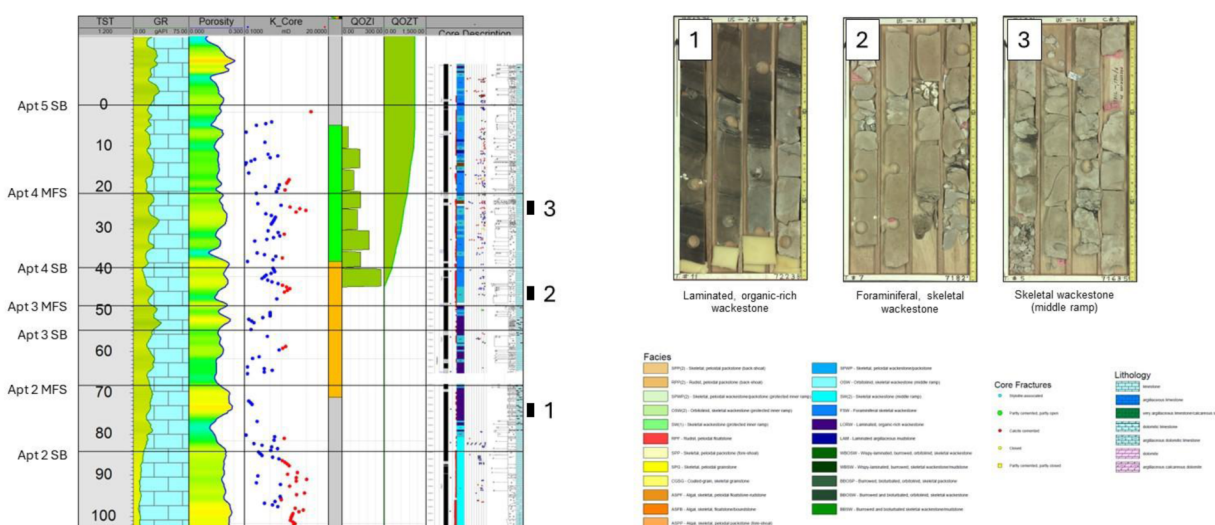


Figure 9—PLT results indicated that the lower section of the Bab member, characterized by organic-rich wackestone with low matrix permeability, did not contribute to the flow, all production originated from the middle section characterised by skeletal wackestone (middle ramp) zone.

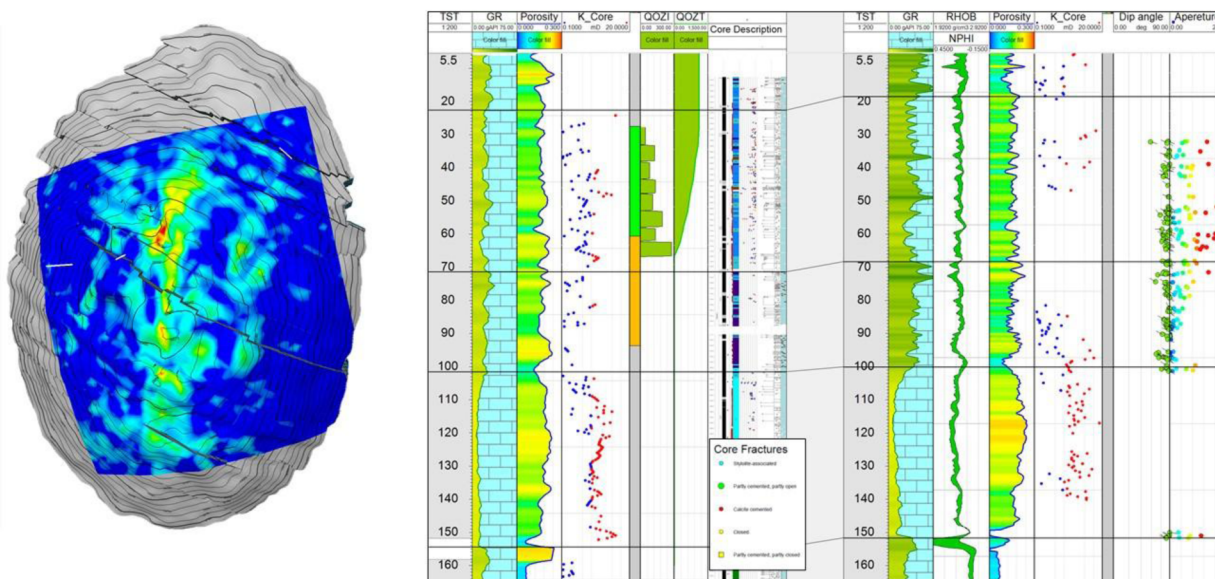


Figure 10—PLT results indicated that the lower section of the Bab member, characterized by minimal fracture aperture, did not contribute to the flow. All production comes from the middle section characterised by higher fracture apertures when compared to the lower section.

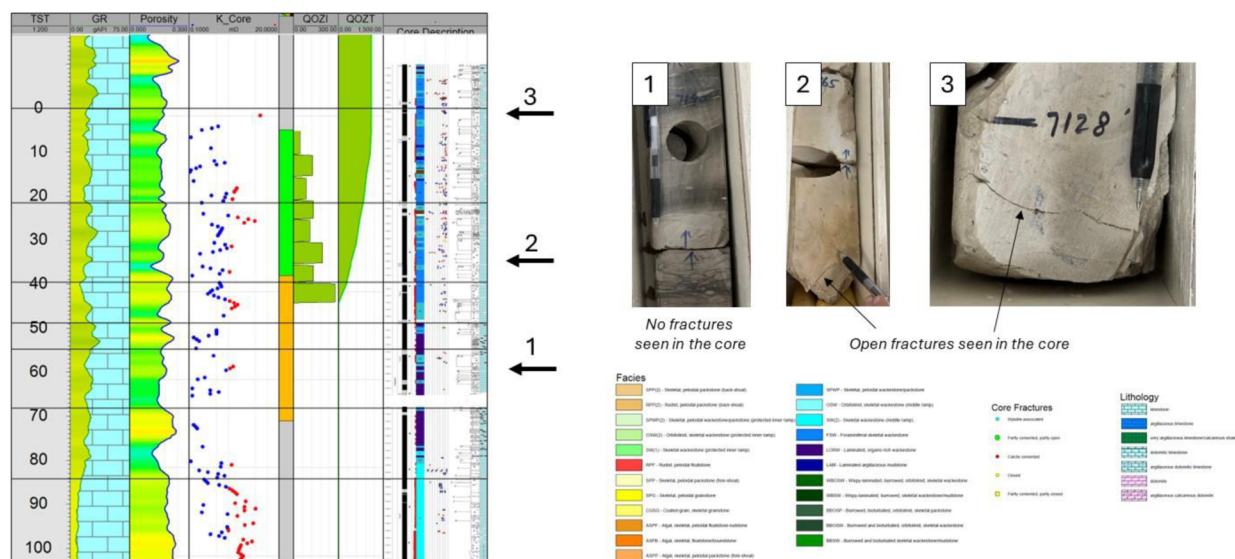


Figure 11—PLT results indicated that despite the low permeability observed in core samples, the middle part of Bab member contributed to production due to the presence of fractures, well-identified and described in image log and core.

Well test interpretation

The detailed review of the well test signatures has provided key insights into the Bab member, allowing for a more accurate characterization. The well test interpretation reveals radial flow stabilizations at conductivity levels reaching several hundreds of millidarcy-feet (mD.ft), significantly higher than the matrix permeability levels, suggesting the presence of fractures Figure 12. However, this permeability enhancement is moderate and, as indicated by the log-log plot analysis, no linear flow regime was identified, which implies that the fractures are relatively small in dimension and sporadically distributed.

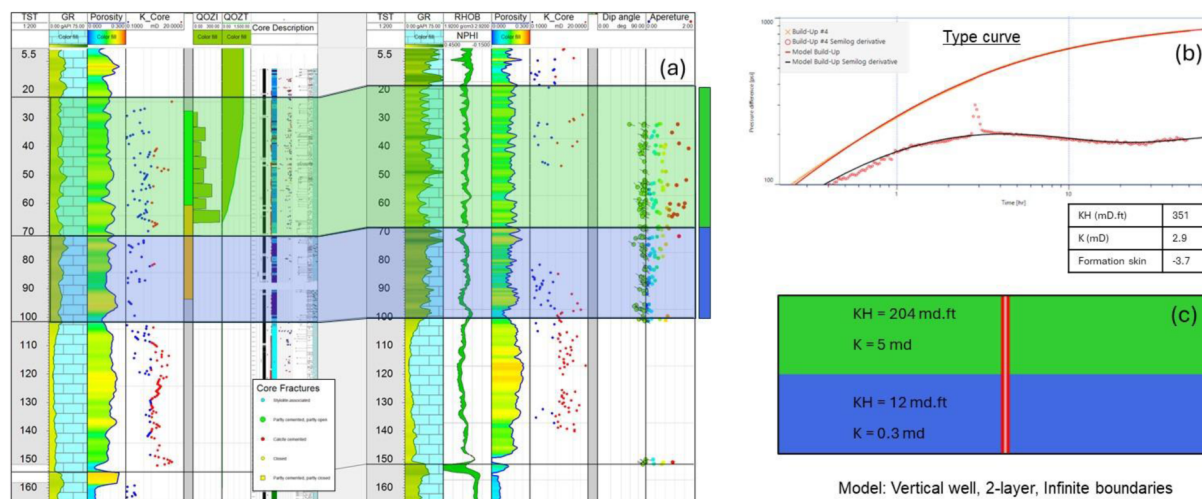


Figure 12—(a) PLT results indicated that the lower section of the Bab member, characterized by minimal fracture aperture, did not contribute to the flow, all production originated from the middle section characterised by higher fracture apertures. (b) The well test interpretation reveals radial flow stabilizations at conductivity levels reaching several hundreds of millidarcy-feet (mD.ft), significantly higher than the matrix permeability levels, suggesting the presence of fractures. (3) A two-layer model used to model the well test response with a vertical well configuration, each two layers exhibited respective permeability-thickness products (KH) of 204 mD.ft for the upper layer and 12 mD.ft for the lower layer.

The well test response was effectively modelled using a two-layer system, each with infinite boundary conditions, and a vertical well configuration. The analysis determined that the two layers exhibited respective permeability-thickness products (KH) of 204 millidarcy-feet (mD.ft) for the upper layer and 12

millidarcy-feet (mD.ft) for the lower layer. These values align precisely with the geological characteristics observed within the Bab member [Figure 12](#). This modelling approach confirms the presence of small-scale fractures that enhance permeability without the presence of significant linear flow features. The integration of sedimentological descriptions and fracture characterization has thus been pivotal in accurately modelling the reservoir, guiding the perforation and stimulation strategies to maximize hydrocarbon recovery from the fractured Bab member.

Discussion

The reassessment and appraisal of the Bab member, a historically low permeability fractured carbonate reservoir within the Thamama group, have shed light on several pivotal aspects. The operation unveiled significant additional reserves, aligning with ADNOC's objective to escalate hydrocarbon production. This discussion will delve into the methodologies employed, the key findings, and the implications for future operations.

The meticulous approach adopted in selecting a well with comprehensive core data and high-quality logs was essential for the operation's success. By re-evaluating the sedimentological description and reservoir properties of the Bab member, potential contributing intervals were pinpointed. The perforation sequence managed to identify these intervals and assess vertical permeabilities effectively, ultimately validating the reservoir's potential.

One of the most striking revelations was the identification of three distinct sedimentological units within the Bab member. The upper and lower units, characterized by laminated, organic-rich wackestone, demonstrated poor reservoir qualities. In contrast, the middle unit, consisting of foraminiferal, skeletal wackestone, emerged as the sole reservoir candidate due to its higher permeability and the presence of open fractures. This finding was pivotal in devising the perforation strategy and optimizing hydrocarbon recovery.

The two-stage perforation strategy underscored the importance of targeting the right intervals. The initial stage aimed at the lower unit and the base of the middle unit, yielding an acceptable production rate. However, the subsequent stage, focusing solely on the middle unit, led to a modest production increment. This indicated high vertical permeabilities, corroborated by PLT data and PBU analysis, which confirmed that production was exclusively from the middle unit while the lower unit acted as a seal.

The operation's success was further validated by PBU analysis, revealing permeabilities five times greater than those suggested by CCA data. This discrepancy hinted at the existence of diffused fractures, substantiated by core descriptions. These fractures significantly enhanced the reservoir's performance, highlighting the critical role of fracture characterization in optimizing hydrocarbon recovery.

Moreover, the implications of this operation extend beyond the Bab member. Paleogeographic mappings suggest that similar characteristics may be present in other offshore fields. This opens up avenues for analogous operations to unlock further potential reservoirs, contributing to ADNOC's goal of accelerating hydrocarbon production.

The reassessment and appraisal of the Bab member have not only unlocked significant additional reserves but also provided valuable insights into the reservoir's characteristics and dynamic behaviour. The success of this operation underscores the importance of meticulous planning, sedimentological and fracture characterization, and targeted perforation strategies in optimizing hydrocarbon recovery from fractured carbonate reservoirs. Future operations can leverage these findings to explore and exploit similar reservoirs, thereby contributing to ADNOC's aspiration to increase hydrocarbon production.

Conclusion

The comprehensive appraisal and reassessment of the Bab member have yielded pivotal insights and uncovered substantial additional reserves, aligning with ADNOC's ambitions to boost hydrocarbon production. By meticulously selecting a well with comprehensive core data and high-quality logs, and

undertaking a thorough re-evaluation of the sedimentological description and reservoir properties, the team effectively pinpointed potential contributing intervals. The strategic perforation sequence, designed in collaboration with operations and reservoir management teams, proved instrumental in identifying these intervals and assessing vertical permeabilities.

The discovery of three distinct sedimentological units within the Bab member—the upper and lower units with laminated, organic-rich wackestone, and the middle unit with foraminiferal, skeletal wackestone—was crucial. The middle unit's higher permeability and presence of open fractures made it the sole viable reservoir candidate, guiding the perforation strategy and optimizing hydrocarbon recovery.

The operation's success was underscored by PLT and PBU analyses, which validated the reservoir's potential and confirmed that production was exclusively from the middle unit while the lower unit acted as a seal. The revelation of permeabilities five times greater than those suggested by CCA data indicated the existence of diffused fractures, significantly enhancing the reservoir's performance.

These findings extend beyond the Bab member, suggesting potential similar characteristics in other offshore fields. This opens avenues for analogous operations to unlock further potential reservoirs, contributing significantly to ADNOC's goal of accelerating hydrocarbon production.

The Bab member reassessment has not only unlocked significant reserves but also provided valuable insights into the reservoir's characteristics and dynamic behaviour. Future operations can leverage these findings to explore and exploit similar reservoirs, advancing ADNOC's aspiration to increase hydrocarbon production.

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